

MODULE 01 · REST API INTERNALS



How Ollama APIs Work

`/api/generate` vs `/api/chat` — Under the Hood

Chat vs. Generate: What is the Difference?

`/api/generate`

- **How it works:** One-time request and response.
- **Input:** A simple text prompt.
- **Memory:** No memory. Every request is fresh.
- **Best for:** Simple tasks like reading data or sorting text.

`/api/chat`

- **How it works:** Back-and-forth conversation.
- **Input:** A list of messages with roles (system, user).
- **Memory:** You must send the history every time.
- **Best for:** Building chatbots and AI agents.

How the `/api/generate` Request Looks

```
{  
  "model": "qwen2.5-coder:7b",  
  "prompt": "Write a Python function to check if a number is prime.",  
  "stream": false,  
  "options": {  
    "temperature": 0.2  
  }  
}
```

How the `/api/chat` Request Looks

```
{
  "model": "qwen2.5-coder:7b",
  "messages": [
    {
      "role": "system",
      "content": "You are a senior code reviewer."
    },
    {
      "role": "user",
      "content": "Is this prime checker function optimal?"
    }
  ],
  "stream": false
}
```

How to Read Performance Numbers

```
{
  "model": "qwen2.5-coder:7b",
  "created_at": "2026-06-29T16:30:00Z",
  "response": "...",
  "done": true,
  "total_duration": 1823450600,
  "load_duration": 1502300,
  "prompt_eval_count": 14,
  "prompt_eval_duration": 125000000,
  "eval_count": 48,
  "eval_duration": 1548000000
}
```

- `load_duration` **Loading Time** — Time taken to load the model into memory.
- `prompt_eval_count / duration` **Reading Speed** — How fast the system reads your prompt.
- `eval_count` **Writing Speed** — How fast the model generates new words.

Common Problems and How to Fix Them

Model Unloads Automatically

Ollama removes models from memory after 5 minutes of no use.

→ Fix: Set `OLLAMA_KEEP_ALIVE=-1` to keep the model loaded forever.

Long Messages Get Cut Off

The default memory size is small. Large chats will lose older messages.

→ Fix: Add `"num_ctx": 8192` inside options.

Requests Wait in Line

Ollama handles requests one by one, slowing down multiple users.

→ Fix: Set env configuration `OLLAMA_NUM_PARALLEL=4` to run together.

Next Up: Streaming Responses

Next Up: Module 02 · Streaming Responses

- What we will cover next:
 - Understanding TTFB and latency bottlenecks
 - Setting `"stream": true` and streaming raw JSON chunks
 - Handling line-by-line streaming in Go
 - Handling Nginx proxy buffering & network drops
- Get ready to fix the latency in your apps!